3. Friday, March 29. Compound statements, logically equivalent statements (2.1, 2.2)

Last Time: Introduced
- logical conectors ($\wedge, \vee, \neg, \rightarrow, \leftrightarrow$),
- Compound statements
- Truth Table

Definition 1: A **Tautology** is a compound statement that is always **True**

Example 2: Let P and Q denote statements. Then

- The statement $P \vee (\neg P)$ is a Tautology

P	$\neg P$	$P \vee (\neg P$
T	F	T
F	T	T

- Also $(P \rightarrow Q) \vee P$ is a Tautology:

P	Q	$P \rightarrow Q$	$(P \rightarrow Q) \vee P$
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

- Other Tautologies include $P \rightarrow P$, $P \leftrightarrow P$, $(P \rightarrow Q) \vee (Q \rightarrow P)$. Exercise: check Truth Tables

Definition 3: A **contradiction** is a compound statement that is always **False**.

Example 4: Let P and Q be statements

1) The statement $P \wedge \neg P$ is a contradiction
2) A more complicated one: $(P \wedge Q) \wedge \neg (P \to Q)$ is a contradiction:

P	Q	$P \wedge Q$	$P \to Q$	$\neg (P \to Q)$	$(P \wedge Q) \wedge \neg (P \to Q)$
T	T	T	T	F	F
T	F	F	F	T	F
F	T	F	T	F	F
F	F	F	T	F	F

3) Also $P \leftrightarrow \neg P$ is a contradiction. Check!

Logically equivalent statements

Motivation: What's the difference between $\neg (P \vee Q)$ and $\neg P \wedge \neg Q$?

P	Q	$P \vee Q$	$\neg (P \vee Q)$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

"It's not the case that either P or Q"

"Neither P nor Q"

Definition 5: Two expressions are called **logically equivalent** if they have the same truth value for all possible combinations of truth values for all variables appearing in the two expressions. Denoted $X \equiv Y$, \equiv

Example 6: Let P and Q be statements. The following expressions are logically equivalent:

1) $P \equiv \neg(\neg P)$.
2) $P \to Q \equiv \neg P \vee Q$

Proof: We just need to verify they have the same truth tables

1)

P	¬P	¬(¬P)
T	F	T
F	T	F

As desired.

2)

P	Q	P → Q	¬P	¬P ∨ Q
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

These expressions have the same truth table, therefore they are equiv.

Example 7: (Negations of conditionals)

We have $\neg(P \to Q) \equiv P \wedge \neg Q$ for any statements P and Q

Proof: Recall from Motivation that $\neg(R \vee S) \equiv \neg R \wedge \neg S$. Then

$$\neg(P \to Q) \overset{\text{Example 6 (2)}}{\equiv} \neg(\neg P \vee Q) \overset{\text{Motivation}}{\equiv} \neg(\neg P) \wedge \neg Q \overset{\text{Example 6 (1)}}{\equiv} P \wedge \neg Q$$

Therefore $\neg(P \to Q) \equiv P \wedge \neg Q$ □

More on Conditionals: For a conditional $P \to Q$, we have three related conditionals

- The converse $Q \to P$
- The inverse $\neg P \to \neg Q$
- The contrapositive $\neg Q \to \neg P$

Next we compute the truth tables

P	Q	$\neg P$	$\neg Q$	$P \to Q$	$Q \to P$	$\neg P \to \neg Q$	$\neg Q \to \neg P$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

Therefore $P \to Q \equiv \neg Q \to \neg P$!

This also justifies the Truth Table for $P \to Q$.

Next we summarize some useful logical equivalences

Theorem: Let P, Q and R be statements. Then

- $\neg(\neg P) \equiv P$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ } DeMorgan's Laws
- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ } Distributive Laws
- $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

- $P \to Q \equiv \neg P \vee Q$
- $\neg(P \to Q) \equiv P \wedge \neg Q$
- $P \to Q \equiv \neg Q \to \neg P$
- $P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$